# Translating identifiers

Converting the identifiers in one column of a loaded table into another identifier system, and writing the result back as a new column.

**AT A GLANCE**

| | |
|---|---|
| Where | Data → ID Translator |
| Data levels | Protein and phosphosite |
| Works on | One column of the uploaded table, chosen by name |
| Systems | Eighteen, from HGNC and UniProt. Any of them can be the input and any of them the output. |
| Source | A lookup table bundled with the application, BioNamesDB.txt: a prepared merge of UniProt and HGNC content, about 107,000 mappings. |
| Writes | A new column on the active dataset, plus a downloadable copy of the whole table. |
| In viewer mode | Not available: the page is not shown, and running or downloading a translation is refused. |

## 1 What it translates

An uploaded report names its features in whatever system the search engine used — a UniProt accession, most often — while the rest of the world asks for gene symbols. This page does that conversion on the table itself: one column in, one new column out, beside the original. Nothing is replaced and no row is dropped, so a translation that fails is visible as an NA in the new column rather than as a shorter table.

Eighteen systems are offered, and each of them can be either side of the translation:

- HGNC: HGNC ID, HGNC Symbol, HGNC Name, HGNC Alias Symbol.
- UniProt: UniProt Accession, UniProt Name, UniProt Gene, UniProt Description, UniProt Secondary Accession, and UniProt All Accessions, which is the primary accession and the secondary ones treated as one set.

- Other cross-references carried in the same table: Ensembl, VEGA, ENA, RefSeq, CCDS, PubMed, Enzyme Commission (EC), RNAcentral.

The mapping is whatever the bundled table holds. It is a snapshot, prepared from UniProt and HGNC downloads and shipped with the application; it is not fetched from anywhere at run time, so translation works with no network and gives the same answer on every machine running the same release. A record absent from that table cannot be translated here however current it is elsewhere.

> **NOTE**
>
> The release the snapshot was built from is not recorded in this version, so the manual cannot state it. Record the CoPYlot version and retain the exported mapping with the analysis. For time-sensitive identifiers, check the current UniProt or HGNC record independently rather than assuming this bundled snapshot is current.

## 2 Translating a column

1. Choose **Dataset level**. The card beside it names the file currently loaded at that level and its size, so it is clear which table is about to be changed.
2. Choose **Column to translate**. Every column of the uploaded table is offered by name, including ones added by an earlier translation.
3. Choose **Translate to**: the system the new column will be written in.
4. Leave **Auto-detect input database from the selected column** on, or turn it off and set **Translate from** by hand.
5. Press **Translate and Preview**. The four cards report which input system was used, which output system, how many rows were translated out of how many, and the name of the appended column; the first fifty rows are shown below them.

Pressing that button also writes the translated table back as the active dataset for that level, which is why the appended column is available to everything downstream. It does not re-run the annotation: the filtered and $\log_2$ matrices already computed still have the columns they had. Generate the annotation again to carry the new column into them.

**Download Translated Table** runs the same translation and gives the whole table as a file without writing anything back.

**Controls on this page**

| Control | What it does | Default | Where |
|---|---|---|---|
| **Dataset level** | Protein or phospho. Only the levels actually uploaded are offered. | Protein | page |
| **Column to translate** | Which column of the uploaded table holds the identifiers. | the first column | page |
| **Translate to** | The output system, one of the eighteen. | HGNC Symbol | page |
| **Translate from** | The input system. Disabled, and ignored, while auto-detection is on. | HGNC Symbol | page |
| **Auto-detect input database from the selected column** | Works the input system out from the values themselves. | on | page |
| **Translate and Preview** | Translates, shows the summary and the first fifty rows, and applies the result to the active dataset. | — | page |
| **Download Translated Table** | The whole translated table as a file. Does not change the active dataset. | — | page |

## 3 How it is computed

#### THE LOOKUP

The bundled table is read from disk once and then held in memory. For the chosen pair of systems an index is built from it: in every row, each value of the input column is paired with every value of the output column, and rows where either side is empty contribute nothing. The index is therefore many-to-many by construction — one identifier may have several translations, and several identifiers may share one.

#### ONE CELL

A cell is split on semicolons, its parts trimmed, and repeats within the cell dropped, so `P12345;P12345;Q9Y6K9` contributes two identifiers. Each is looked up, and the union of everything found is sorted and written back into one cell, joined by semicolons. A cell that is empty, or whose identifiers are all absent from the index, becomes the literal string `NA`. Translated Rows on the summary counts the rows that are *not* that

string, so it is the number of rows with at least one translation, not the number of identifiers matched.

**AUTO-DETECTION**

Up to 5000 non-empty values are taken from the chosen column and split into at most 20,000 identifiers. For each of the eighteen systems the number of those identifiers that appear anywhere in that system's column of the lookup table is counted, and the system with the highest count wins. A tie is resolved in favour of UniProt All Accessions if it is among the leaders, and otherwise by the order the systems are listed in.
If nothing matches at all and no input system was set by hand, the translation stops and says so. If nothing matches but **Translate from** had been set before auto-detection was switched on, that manual choice is used and a warning records it.

**EASY TO GET WRONG**

Auto-detection overrides **Translate from** whenever it succeeds, so a manual choice made with the checkbox left on is not what runs. Turn the checkbox off to hold the input system fixed — which is what to do when a column contains identifiers of two systems, since detection then sees only the more common one.

## The appended column

The new column is named after the output system, not after the source column, so translating twice into the same system overwrites the first result rather than accumulating columns.

**Column names written, by output system**

| Column | What it holds |
|---|---|
| GeneNames | HGNC Symbol — and also UniProt Gene, which writes the same column name |
| HGNCIDNames | HGNC ID |
| HGNCNames | HGNC Name |
| HGNCAliasSymbolNames | HGNC Alias Symbol |
| UniProtAccessionNames, UniProtNames, UniProtDescriptionNames, UniProtSecondaryAccessionNames, UniProtAllAccessionsNames | the five remaining UniProt systems |

Column names written, by output system (continued)

| Column | What it holds |
|---|---|
| EnsemblNames, VEGANames, ENANames, RefSeqNames, CCDSNames, PubMedNames, ECNames, RNAcentralNames | the cross-reference systems, in the same pattern |

One further column can appear unasked. If the table has no GeneNames column and the output system is not one of the two that write it, the same identifiers are translated a second time into HGNC symbols and added as GeneNames — provided at least one of them maps — so that the analyses keyed on gene names can run. A warning on the summary says when this happened.

### Edge cases

- An identifier that maps to several targets keeps all of them in one cell, separated by semicolons and sorted. Downstream pages that expect one name per row see the whole string.
- An identifier that maps to none leaves NA, and the row stays in the table.
- If the bundled lookup table cannot be found, the page reports that and the locations it searched are written to the server log.

## 4 Exporting

The download is the whole table with the new column, not just the mapping: a .tsv or .txt upload comes back tab-separated with the same extension, everything else comes back as CSV, and the file is named after the upload with _translated before the extension. It is generated from the current settings on the page, so it can be taken without applying anything to the session.

**Behind this page**

| Function | What it produces |
|---|---|
| run_id_translation | Translates and applies the result to the active dataset |
| export_id_translation | The same translation as a file, applying nothing |
| translated_dataframe | The translated table, and the input system that was used |

### Behind this page

| Function | What it produces |
|---|---|
| autodetect_input_db | Works the input system out from the values |
| build_index | The many-to-many index between two systems |
| translate_cell | One cell: split, look up, join, or NA |